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import java.io.*;
import java.util.ArrayList;

public class GradeAnalyzer {
    private static int invalidLines;

    public static void main(String[] args) {
        // Step 1: read scores from file
        ArrayList<Integer> scores = readScores("scores.txt");

        // Step 2: calculate statistics
        double avg = calculateAverage(scores);
        int high = scores.isEmpty() ? 0 : scores.get(0);
        int low = high;
        for (int score : scores) {
            high = Math.max(high, score);
            low = Math.min(low, score);
        }
        // Step 3: write and print report
        writeReport(scores, avg, high, low, "grade_report.txt");
        // Step 4: count grade bands
        countGradeBands(scores);
    }

    // Returns a list of valid scores read from the file
    public static ArrayList<Integer> readScores(String filename) {
        ArrayList<Integer> scores = new ArrayList<>();
        invalidLines = 0;

        try (BufferedReader reader = new BufferedReader(new
        FileReader(filename))) {
            String line;

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        while ((line = reader.readLine()) != null) {
            try {
                int score = Integer.parseInt(line.trim());
                if (score >= 0 && score <= 100) {
                    scores.add(score);
                }
            } catch (NumberFormatException ignored) {
                // count the blank lines and catch invalid scores count
and report in
                if (!line.trim().isEmpty()) {
                    invalidLines++;
                }
            }
        }
    } catch (IOException e) {
        System.err.println("Unable to read " + filename + ": " +
e.getMessage());
    }
    return scores;
}

// Returns the average of a list of scores, or 0.0 if the list is empty
public static double calculateAverage(ArrayList<Integer> scores) {
    if (scores.isEmpty()) {
        return 0.0;
    }
    int total = 0;
    for (int score : scores) {
        total += score;
    }
}

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        return (double) total / scores.size();
    }

    // Writes and prints the report
    public static void writeReport(ArrayList<Integer> scores,
                                   double avg, int high, int low,
                                   String outputFile) {

        String report = "==== Grade Analysis Report ====" +
            System.lineSeparator()
            + "Total Scores Processed: " + scores.size() +
            System.lineSeparator()
            + "Invalid Lines Skipped: " + invalidLines +
            System.lineSeparator()

            //how to add a blank line in the report
            + System.lineSeparator()

            + "Average score: " + avg + System.lineSeparator()
            + "Highest score: " + high + System.lineSeparator()
            + "Lowest score: " + low

            + System.lineSeparator();

        System.out.println(report);

        try (PrintWriter writer = new PrintWriter(new
            FileWriter(outputFile))) {

            writer.println(report);

        } catch (IOException e) {

            System.err.println("Unable to write " + outputFile + ": " +
                e.getMessage());

        }

    }

    //Class to count the grades
    public static void countGradeBands(ArrayList<Integer> scores) {

        int countA = 0, countB = 0, countC = 0, countD = 0, countF = 0;

        for (int score : scores) {

            if (score >= 90) {

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        countA++;
    } else if (score >= 80) {
        countB++;
    } else if (score >= 70) {
        countC++;
    } else if (score >= 60) {
        countD++;
    } else {
        countF++;
    }
}

System.out.println("Grade distribution:");
System.out.println("A (90-100): " + countA);
System.out.println("B (80-89): " + countB);
System.out.println("C (70-79): " + countC);
System.out.println("D (60-69): " + countD);
System.out.println("F (below 60): " + countF);
}

}

```